

# Witness-CL: Evidence-Preserving Online Learning and the Limits of Greedy Probing

Samuel Mausberg

AI-assisted research draft; author review and independent replication pending

## Abstract

Continual improvement requires learning which past evidence changes future behavior while retaining useful prior decisions. Witness-CL is a finite, deterministic reference that learns compatible hidden models from executed traces, trains an online proposer, and admits immutable programs only after baseline-relative checking. We extend the supplied v0.4 artifact with actual Lean verification, a falsifiable decision-regret exploration rule, a local-model inference pilot, and native benchmark interface work. Lean 4.19.0 accepts all 50 inherited statements after proof-script repairs and a new abstract information-obstruction lemma, with no custom axioms or proof placeholders. In a frozen 100-seed, 19,200-episode study, the new probe heuristic reduces mean return by 0.0946 relative to v4 (95% paired interval  $[-0.1692, -0.0200]$ ), while increasing measured mechanics cost. An executable counterexample explains one failure: individually zero-value free probes can jointly reveal a profitable decision, but greedy regret reduction refuses the first probe. We preserve this negative result and specify bounded adaptive probe plans as a falsifiable successor. A real quantized local model executes typed proposals, without a native benchmark or powered superiority claim. Formal guarantees remain about conditional abstract reset-value retention and prepaid deficits; they do not verify the Python implementation, model-class validity, general no-forgetting, or scalable deployment.

## 1 Problem and the actual advance

CL-Bench evaluates whether frontier language agents improve across episodes that share learnable latent structure. Its strongest aggregate system uses raw-history in-context learning (ICL); this does not imply that every memory method loses on every metric or backbone. The paper reports immediate-observation overfitting and weak reuse across instances [1]. AgentCL uses controlled compositional, naive, and held-out streams to distinguish reuse from mere persistence; irrelevant memory can degrade performance [2]. These results motivate a more exact question: what part of experience should a learner treat as state, and what evidence warrants letting that

state change its future behavior?

The open target is a cost-effective, deployed learner that improves across useful stateful tasks from its own episodes without an offline retraining phase and without losing valuable prior behavior. Merely preserving weights, transcripts, or a skill file does not imply preserving downstream competence. Conversely, not every episode can be required to improve if useful information sometimes requires a costly experiment. A meaningful solution must separate retained competence from exploration loss and account for both.

**Inherited mechanism.** The v0.3 ancestor of the supplied v0.4 bundle provides a finite list of possible worlds, observable complete state, continuation-aware improvement, incumbent retention, and prepaid exploration. Its synthetic late return matches rather than beats a strong full-history learner. Separate noisy-evidence and frozen-neural-module branches do not solve representation acquisition. We preserve that Git history and archive its paper and results rather than relabel them as new experiments.

**Version 4.** We replace the supplied world list and observed latent state with a learned set of hidden transducers consistent with executed traces. Only a structural class bound, reset semantics, finite alphabets, and public reward functions are supplied. The same loop now contains online neural proposal updates, exact model-set conditioning, immutable observation-contingent programs, paired continuation verification, and informative prepaid probes. No hidden world table, state, or regime label is given to the learner.

The advance is deliberately restricted. Discovering compatible latent models inside a correctly declared finite class is not discovering a sufficient abstraction of arbitrary language or computer-use traces. Our proposed high-impact direction is to learn affordable continuation contracts: compact predictive descriptions sufficient to justify useful behavioral changes. The reference makes this direction executable and falsifiable. Its current empirical result is not benchmark superiority.

**Version 5 contribution and result.** We audit the v0.4 bundle rather than assume its claims are verified. We compile and repair its abstract Lean proofs, test

decision-directed exploration with a frozen rule, diagnose its failure, execute local-model inference, and pin the native benchmark interface. The new exploration rule loses on the primary held-out metric. Its failed test and complementary-probe counterexample constrain the successor design; they are not presented as a solution to continual learning. The original v0.4 paper and data remain archived unchanged.

## 2 Prior work constrains the claim

**Memory and executable experience.** ACE develops evolving playbooks; ALMA meta-learns memory designs; EvoLib builds and consolidates reusable knowledge abstractions [3, 6, 12]. These are substantive adaptation mechanisms, not simply retrieval stores. ALMA’s outer search also makes it important to distinguish prior design cost from strictly online adaptation. None of these comparisons has been reproduced here.

EvoTest evolves system configurations between episodes, while TTHE evolves the executable harness and commits changes using execution-derived proxy signals [4, 5]. TTHE’s main evaluation adapts and scores on the same batch; its authors identify next-batch evaluation and stronger compute matching as open questions. A September 4, 2026 study constructs persistent versioned skills for computer-use agents, freezes the library during each iteration, and reports benefits over a configuration-matched empty-library control along with revision failures [13]. Versioned libraries and frozen execution snapshots are therefore not new contributions of Witness-CL. The unresolved distinction is when evidence supports a claim about later behavior, rather than only a favorable proxy or recent task.

**Latent state and conservative learning.** Predictive state representations already define state through predictions of future observable experiments [10]. Dynamic shielding learns an approximate automaton from black-box interaction and filters potentially unsafe exploration [11]. DeepSPI already combines online safe policy improvement, learned world models, and representations, with conditional theory and ALE-57 evaluation [8]. Its learned improvement mechanism is a direct antecedent; abstraction validity and the assumptions in its bounds remain substantive requirements. These are direct precedents, not peripheral related work. Our exact deterministic class is narrower than those applications.

Conservative exploration methods such as StepMix address explicit baseline performance constraints [9]. The prepaid debit argument here is not the first conservative-exploration theorem. Progressive networks already isolate previous learned components [7]; neither freezing old functions nor changing parameters only online is novel by itself. We claim an implemented combination

with explicit failure tests, not established novelty of the combination or a new universal learning principle.

## 3 A precise latent-learning contract

Let  $A$  and  $O$  be known finite action and output alphabets. The unknown world  $m^*$  is a stationary deterministic transducer with at most  $K$  states, embedded in  $S = \{0, \dots, K-1\}$ . A complete table specifies

$$T_m : S \times A \longrightarrow S \times O. \quad (1)$$

Every episode begins at the same actual reset state 0. Hidden state labels and the table are never observed. The learner receives the executed trace  $\tau = ((a_0, o_0), \dots, (a_{H-1}, o_{H-1}))$  at finite horizon  $H$ . There is no stochastic feedback in the integrated experiment.

A public objective  $g$  specifies a known bounded integer reward  $r_g(o)$ . A policy  $p$  is a complete immutable tree:  $p_t : O^t \rightarrow A$ . The reset return is

$$V_m(p, g) = \sum_{t=0}^{H-1} r_g(o_t), \quad a_t = p_t(o_{<t}). \quad (2)$$

A protected incumbent  $b_{n,g}$  is stored for each declared objective. The objective identifies the requested utility, not a hidden dynamics regime. The common initial anchor is  $b_{0,g}$ .

Let  $\mathcal{D}_n$  contain all actually executed traces before episode  $n$ . Define

$$\mathcal{C}_n = \{m \in \mathcal{M}_K : \forall \tau \in \mathcal{D}_n, m \models \tau\}. \quad (3)$$

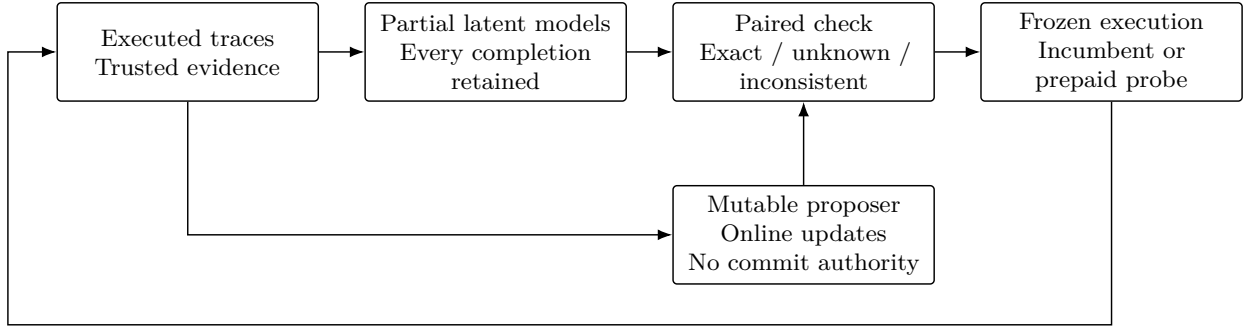
Here  $m \models \tau$  means that the table realizes the observed trace. The model class has  $(K|O|)^{K|A|}$  labelled complete tables. Supplying  $K$  is a substantial prior. Data consistency cannot prove that  $m^* \in \mathcal{M}_K$ . Theorems below explicitly assume this coverage and unchanging reset semantics.

**Four different claims.** Preserving raw evidence is not the same as preserving a learned function. Preserving that function is not the same as preserving a routed incumbent’s reset value. Preserving incumbent value does not require every exploratory episode to attain that value. We prove the third property under this contract and separately bound the fourth property’s possible deficit. We do not prove that the mutable neural network itself avoids forgetting.

## 4 Learning an exact partial-model cover

A partial table  $c$  assigns some state–action edges and leaves the rest undefined. Its denotation is

$$[c] = \{m \in \mathcal{M}_K : c \subseteq T_m\}, \quad (4)$$



**Figure 1:** Implemented loop. Model rollouts do not become observed training labels. The proposer may forget internally; only checked immutable programs replace a protected incumbent. All guarantees are conditional on a correct finite class, stationary dynamics, declared rewards, and genuine resets.

where inclusion means agreement on every assigned edge. A finite frontier  $\mathcal{F}_n$  represents  $\mathcal{C}_n = \bigcup_{c \in \mathcal{F}_n} [c]$ . Initially the single all-undefined table represents the entire class. No complete list of worlds is passed to the learner.

To process one reset trace, pair every frontier table with state zero. For observed  $(a, o)$ , retain a defined edge only when its output is  $o$ . For an undefined edge, branch over all  $K$  successor states, assigning output  $o$  in every branch. Retain the branch’s hidden successor while processing the trace. Merge duplicate table–state pairs. At episode end discard the end state, since the next episode resets, and merge duplicate tables. Undefined edges are not filled with a likely default during inference.

**Theorem 1** (Evidence-complete cover). *With exact feedback and no search truncation, the update above represents exactly  $\mathcal{C}_n$ . If  $m^* \in \mathcal{M}_K$ , every updated cover contains  $m^*$ .*

*Proof.* Induct on observations within each reset trace. For any compatible complete table, a defined visited edge must have the observed output; an undefined edge has its true successor among the  $K$  branches. Thus no compatible table is lost. Conversely, every retained branch assigns the observed output to the visited edge and tracks its successor, so all its completions realize the processed prefix. Resetting the branch state between episodes is valid by assumption. Deduplication changes representation, not its union of completions. Induction over episodes gives equality; actual feedback gives truth membership.  $\square$

This cover can overlap and can grow exponentially. The implementation caps the frontier and marks incomplete inference rather than silently retaining only a likely subset. Once coverage is unknown, new certificates are not issued. An empty exact class is reported as INCONSISTENT, never as vacuous success.

**Changing capacity.** Appending a new candidate program does not change  $\mathcal{C}_n$ . Increasing  $K$  does. The implementation replays all raw evidence under the new

bound, starts a new contract era from the anchor, and does not inherit old certificates or the old budget claim. This is an explicit online recovery operation, not evidence that a fitted smaller model was sound before expansion.

## 5 Checking a pair rather than trusting a model

For a candidate  $p$  and current incumbent  $b$ , define

$$L_n(p, b, g) = \min_{m \in \mathcal{C}_n} [V_m(p, g) - V_m(b, g)], \quad (5)$$

$$U_n(p, b, g) = \max_{m \in \mathcal{C}_n} [V_m(p, g) - V_m(b, g)]. \quad (6)$$

Computing these extrema separately by materializing every complete world is unnecessary when many transitions cannot affect the comparison.

The checker symbolically executes both programs in the *same* partial table. It expands an undefined edge only when one execution reads it, branching over its  $K|O|$  possible assignments. Both executions share each assignment. It accumulates their reward difference and retains a complete witness for the minimum. Unvisited slots remain universally quantified.

**Closing a suffix.** At the same time index, if both executions have the same hidden state and identical remaining observation-contingent programs, their remaining reward difference is zero for every completion. This follows by induction on future steps. Equal current observations alone do not justify closure: hidden states with the same emitted symbol can have different successors.

We compile remaining program trees into exact pair-local directed acyclic graph (DAG) IDs. The interning key is the complete tuple consisting of the current action and child IDs. Dictionary equality checks the whole tuple, so a hash collision cannot equate different programs. This is not a shapes-only cache or a learned equivalence prediction. DAG construction cost is included in timings.

**Theorem 2** (Exact paired extrema). *For a nonempty exact cover, a completed paired search returns precisely (5) and (6). A returned lower-bound witness is a compatible complete table attaining the minimum.*

*Proof.* Expanding an unread edge partitions its completions into all possible assigned edges. A defined edge needs no split. Therefore every compatible world follows a search branch, and no branch introduces an incompatible visited edge. At the horizon the accumulated difference is constant over all unread completions. The same holds at an early closure because subsequent states, programs, observations, and rewards agree inductively. The minimum and maximum of these leaf differences are consequently the desired extrema. Any remaining undefined slots at a minimizing leaf may be completed arbitrarily, producing a compatible witness with the same difference.  $\square$

A node cap yields UNKNOWN with no usable bound. It does not turn visited leaves into a confidence interval. For each partial, at most  $u = \min(K|A|, 2H)$  distinct unresolved slots are read along a paired path, giving up to  $(K|O|)^u$  assignments, in addition to horizon and frontier overhead. Full policy-tree construction itself costs  $O(\sum_{t=0}^{H-1} |O|^t)$ . Conditional sharing is not polynomial-time learning.

**Corollary 3** (Maximality for the declared class). *For a fixed candidate and incumbent,  $L_n \geq 0$  is necessary and sufficient for nonnegative return difference in every world of  $\mathcal{C}_n$ . If  $L_n < 0$ , its witness refutes uniform admission within this contract.*

*Proof.* Sufficiency is the definition of a minimum. Necessity follows because the finite nonempty class attains that minimum. Theorem 2 supplies a witness.  $\square$

This does not make the overall learner optimal. A different experiment, candidate, objective, or weaker guarantee may yield higher return.

## 6 Online training and informative exploration

### 6.1 The proposer has no admission authority

The reference trains a small tanh network after every executed episode. Features combine the public objective with per-time action frequencies of a candidate program. With normalized observed return  $y_n$ , the loss is

$$\ell_n(\theta) = \frac{1}{2}(f_\theta(g_n, p_n) - y_n)^2. \quad (7)$$

One stochastic gradient update uses the selected program and its actual return. There are no evaluator-oracle labels, imaginary rollout labels, or offline optimization

stages. A visit bonus encourages underexplored proposals. These features collapse distinct reactive trees and are intentionally weak; the primary library contains only open-loop words.

Every proposal is a typed immutable program. A candidate replaces the incumbent only after an exact check proves strictly positive  $L_n$ ; ties preserve the incumbent. An optional local-model driver accepts bounded JSON words or trees, not executable code or self-issued certificates. Its parser is tested, but no real LLM service ran in v4. The separate v5 pilot below closes that execution gap without constituting a native benchmark adapter.

**Theorem 4** (Proposer-independent retention). *Within a stationary realizable era, suppose every incumbent replacement is an immutable program with a completed check satisfying  $L_n \geq 0$ . Then for each declared objective  $g$ , the true reset values of its successive incumbents are nondecreasing, regardless of how proposals are generated or updated.*

*Proof.* Theorem 1 retains  $m^* \in \mathcal{C}_n$ . Substituting it into the universal bound gives  $V_{m^*}(p, g) \geq V_{m^*}(b, g)$ . Induct over admitted replacements. Proposal quality does not occur in the inequality, only in which candidate is checked. Frozen program semantics and unchanged objective/reset semantics are necessary for the substitution.  $\square$

### 6.2 A non-refundable exploration ledger

For a non-admitted probe  $p_n$ , charge before execution

$$d_n = [-L_n(p_n, b_{n,g_n}, g_n)]_+. \quad (8)$$

Only a completed exact comparison can authorize a probe, and cumulative debits must remain at most  $B$ . Favorable realized outcomes do not refund debits. Incumbent execution and admitted improvements have zero debit.

**Theorem 5** (Completed-prefix deficit). *Under Theorem 4’s assumptions, every completed episode prefix  $N$  satisfies*

$$\sum_{n=1}^N [V_{m^*}(b_{0,g_n}, g_n) - R_n] \leq \sum_{n=1}^N d_n \leq B. \quad (9)$$

*Proof.* Retention gives  $V_{m^*}(b_{0,g_n}, g_n) \leq V_{m^*}(b_{n,g_n}, g_n)$ . Exact debit validity bounds the latter by  $R_n + d_n$  for each executed probe; the same relation holds with  $d_n = 0$  for incumbent or admitted execution. Summation and the prepaid ledger yield the claim for every completed prefix.  $\square$

Equation (9) allows individual losses. It is not an all-state performance guarantee, a guarantee under hidden dynamics changes, or a safety specification for irreversible actions. It applies separately within each contract era, not across unbounded capacity restarts.

**Algorithm 1. One reset episode of the reference learner**

1. Rank immutable programs with the online proposer. Compare each candidate with the selected public objective’s incumbent using exact paired execution.
2. If a completed check has  $L > 0$ , promote a candidate with maximal  $L$ , breaking ties by proposer order. Execute it without a debit.
3. Otherwise consider only exact checks with  $U > 0$  and affordable  $[-L]_+$ . Enumerate observable traces. If an informative probe is available, select the least debit, then most traces, then largest  $U$ , then proposer order. Charge its debit before execution. Otherwise execute the incumbent.
4. Condition the cover on the actual trace. Apply one neural update using only the selected program and observed return. No hypothetical outcome is evidence.

**Figure 2:** The preserved v4 loop. Version 5 changes only probe selection; certificate admission, immutable execution, evidence updates, and prepaid debits are retained.

### 6.3 Do not pay for a certainly uninformative probe

Our initial integrated controller could spend repeated debits on programs whose observable outcome was already fixed. We now enumerate a feasible probe’s set of distinct observable traces over  $\mathcal{C}_n$ . A probe is selected only when this set contains more than one trace and  $U_n > 0$ .

**Proposition 6** (Strict evidence reduction). *If a deterministic probe has at least two possible observable traces in  $\mathcal{C}_n$ , every realized trace removes at least one currently compatible world.*

*Proof.* Two different possible traces cannot both equal the observed trace. A compatible world realizing the other trace is eliminated by conditioning.  $\square$

This is not expected information gain. The controller orders feasible probes by smallest debit, most distinct observable traces, largest optimistic return difference, then proposer rank. Counts are not probabilities and do not count arbitrary latent-state labelings as information. At most  $|\mathcal{C}_0| - 1$  strictly eliminating events are possible in a finite class, but this exponential bound does not guarantee sufficient budget, available distinguishing programs, or identification of an optimal policy.

## 7 Inherited v4 experiments, reproduced checks

### 7.1 Protocol and comparators

Development uses seeds 0–19. An initial seven-arm run contains 6,720 episodes; the revised eight-arm development run contains 7,680. A local protocol record was written before generating 100 fresh seeds, 10000–10099. It is a local freeze, not external preregistration. The primary run comprises eight systems, 48 episodes per seed, and 38,400 episode rows.

Each hidden transducer has two states, two actions, and two outputs. The horizon is four, and all 16 binary action words form the candidate library. The learner does not receive the generated table. Public rewards alternate in three 16-episode blocks:  $r_A = (0, 2)$ ,  $r_B = (2, 0)$ , then  $r_A$  again. The transition model never changes. This is public-objective recurrence, not hidden-regime recurrence. Episode rewards lie in  $[0, 8]$ .

The strong full-history comparator enumerates all consistent models and selects the word with largest optimistic value, breaking ties by mean model value and then lexicographic order. It has no risk constraint. It is a symbolic planner, *not* raw-history LLM ICL. A last-output model is a weak observation-reactive diagnostic. Other arms are the fixed anchor, Witness at budgets 0, 16, and 64, a no-training ablation, and an unguided-probe ablation.

All systems receive only their own executed feedback and have equal environment horizons. Total CPU time is not matched. Evaluator-only word oracles do not train the agents. The primary metric is mean return over all episodes; late-A return uses the last eight. We additionally track incumbent regressions, risk spent, and maximum completed-prefix deficit against the fixed anchor.

**Held-out limitations and audit.** The 256-table class is tiny. Development contains 19 distinct tables and the 100 fresh seeds contain 81, with six tables shared between the two sets. Independent fresh seeds therefore do not establish unseen-world generalization. A secondary new-reward diagnostic in raw JSON preserves objective A’s output ordering and is not treated as credible new-domain transfer.

After the first held-out run, exact DAG IDs replaced repeated structural suffix checks and strict candidate registration was added. The controller’s selection rule was not tuned on held-out outcomes. The primary run was repeated with the final code. An included audit verifies identical actions/programs, rewards, objectives, deficits, spent risk, and frontier sizes across all 38,400 rows. Original freeze hashes and pre-backend results are retained; timings and search counters are allowed to differ.

| System                       | Mean   | Late A | Max. deficit |
|------------------------------|--------|--------|--------------|
| Fixed anchor                 | 3.9267 | 3.7800 | 0            |
| Full-history planner         | 6.3729 | 6.4800 | 2            |
| Last-output model            | 6.0275 | 6.1450 | 52           |
| Witness, $B = 0$             | 6.2354 | 6.4000 | 0            |
| Witness, $B = 16$            | 6.3358 | 6.4800 | 4            |
| Witness, $B = 64$            | 6.3358 | 6.4800 | 4            |
| No neural training, $B = 64$ | 6.3217 | 6.4800 | 4            |
| Unguided probes, $B = 64$    | 6.2792 | 6.4800 | 40           |

**Table 1:** Executed primary study: 100 fresh seeds, eight systems, 48 episodes, 38,400 rows. Mean and late-A values are rewards, not percentages. Maximum deficit is the worst completed-prefix anchor deficit across all seeds. The strong full-history planner is symbolic, not an LLM ICL baseline.

## 7.2 Learning and retention results

Table 1 shows improvement over the fixed anchor, but not over the strong planner. Guided budgets 16 and 64 have mean return 6.3358 versus 6.3729. The paired-seed difference is  $-0.037083$  reward per episode, with a 95% paired  $t$  interval of  $[-0.055284, -0.018883]$ . Average-return superiority is rejected in this experiment. Late-A return ties at 6.48. The planner also has a lower largest observed prefix deficit, 2 versus 4. The conditional bound is not evidence of an empirical risk–return advantage over this planner.

The informative-probe refinement is useful within this controller. Guided B64 exceeds unguided B64 by 0.056667 reward per episode, interval  $[0.029805, 0.083528]$ , and reduces the largest observed prefix deficit from 40 to 4. Training exceeds the no-training B64 ablation by only 0.014167, interval  $[0.007394, 0.020940]$ . These secondary comparisons are descriptive and not multiplicity-corrected confirmatory findings. The small neural effect argues against presenting this as a major new training algorithm.

There are zero recorded incumbent regressions under the realizable contract. Mean B16 risk spent is 2.62, versus the allowed 16; mean incumbent promotions are 1.4 per seed. The network receives 48 actual-return updates per seed. Its weight payload is 1,544 bytes, but this excludes features, visits, trees, evidence, frontiers, archives and Python overhead. Raw trace payload grows with episode count. No bounded-total-memory result follows.

## 7.3 Checker cost and robustness

A separate 60-query CPU diagnostic compares paired execution with and without suffix closure, using seven timing repetitions per query and identical extrema. Median query speedups are 8.41 for identical programs, 1.55 for shared suffixes, and 0.90 for unrelated programs. The identical-program case is a trivial short-circuit, not an end-to-end systems claim. Unrelated programs remain slower. Before DAG IDs, repeated subtree work erased

the shared-suffix benefit; those negative measurements are preserved.

A 15-case scaling stress test varies state count and horizon with a 20,000-node cap. Four cases return UNKNOWN. This is correct abstention but unfavorable scaling, not a hidden success. The reference has no new GPU kernel, and its CPU microbenchmarks do not predict GH200 serving performance.

An additional 1,200-episode diagnostic overwrites proposer parameters with large arbitrary values between episodes. Incumbent and budget violations remain zero under the contract. This tests separation of proposal and admission, not adversarial safety of the whole Python process. An unannounced dynamics-change counterexample has old certified gain +2 and actual new-world difference  $-2$ . Feedback can detect some contradictions after execution; it cannot protect the first violation of an unobserved assumption.

## 8 Implementation and proof status

The release adds an exact partial-table learner, paired checker, online agent, strict proposal parser, three experiment drivers, independent C++ rollout reference, and tests. All 355 inherited Python tests reproduce; the v5 suite adds mechanism, local-pilot, and native-interface checks. Exact small-class tests compare model covers and paired extrema against exhaustive enumeration, including reactive policies, lower-bound witnesses, empty classes, search caps, capacity changes, and wrong-state-bound counterexamples. The neural gradient is checked by finite differences.

The new C++ reference passes 400 paired observation-tree cases and a bounded reward rejection under undefined-behavior sanitization. The preserved legacy suite passes 632 cases plus an overflow check. These tests validate complete numeric rollouts and boundary handling, not a formal refinement of the symbolic search. The inherited v4 release did not run a native benchmark,

actual LLM, or CUDA experiment. Version 5 adds actual local-model inference as described below; no new CUDA kernel or native performance result is claimed.

Lean 4.19.0 now accepts all 50 inherited declarations and one new history-aliasing obstruction. Two reserved binder names and one Boolean proof step required repair; no inherited hypothesis or conclusion was weakened. An automated inventory prints every theorem’s axioms and ties the report to source hashes. Twenty-four declarations are axiom-free; the remainder use only Lean’s standard propositional extensionality, quotient soundness, and classical choice. No custom axioms or final `sorryAx` dependencies remain.

This is verification of the encoded abstract statements. For example, a cover splitting lemma does not prove the Python depth-first search implements that split, and a guarded-update lemma assumes a correct bound and true-model membership. Numerical training, search-cap propagation, runtime tickets, C++, and LLM semantics are not formally refined to these assumptions. The source-hashed audit and detailed proof-to-runtime gaps accompany the code.

**Proposition 7** (History aliasing prevents uniform strict gain). *Suppose two worlds give a deterministic policy identical available history, including its memory. If every action that strictly improves on an incumbent in the favorable world loses value in the adverse world, a policy preserving the incumbent’s value in the adverse world cannot improve strictly in the favorable world at this information state.*

*Proof.* Identical histories give the same selected action. Strict favorable gain would, by hypothesis, make that action worse than the adverse incumbent, contradicting adverse preservation. The Lean statement instantiates this implication for arbitrary history and action types with integer value functions.  $\square$

This elementary obstruction does not forbid safe distinguishing probes, nor claim a new lower bound for randomized or noisy learning. It makes the scope of universal zero-loss discovery explicit.

## 9 Version 5: a falsified exploration hypothesis

### 9.1 Decision regret instead of trace count

The inherited controller charges a worst-case loss but orders feasible probes by how many observable traces they distinguish. This can reward distinctions that do not change which program should be executed. We tested a replacement that values uncertainty about the

decision itself. For the fixed program library  $\mathcal{P}$ , define

$$\rho_g(C) = \min_{p \in \mathcal{P}} \max_{m \in \mathcal{C}} \left[ \max_{q \in \mathcal{P}} V_m(q, g) - V_m(p, g) \right]. \quad (10)$$

This is minimax simple regret restricted to the supplied library, not realized cumulative regret or unrestricted optimal control. Model-wise optima are computed inside the learner’s compatible class; the true environment is not revealed. Models with identical observable traces for every library program are quotiented before scoring. Label multiplicity is therefore not treated as probability.

Let  $\mathcal{C}_{u,\tau}$  be the nonempty cell consistent with observable trace  $\tau$  after program  $u$ . The fixed two-objective score is

$$G(u, \mathcal{C}) = \sum_g \rho_g(C) - \max_{\tau} \sum_g \rho_g(\mathcal{C}_{u,\tau}). \quad (11)$$

Among exactly checked, affordable probes, the new controller maximizes  $G/(1+d)$ , then favors lower debit, larger optimistic current-objective gain, and proposer order. The incumbent is eligible as a free probe. If the best  $G$  is zero, the controller executes the incumbent. Strict guaranteed improvements still use the unchanged v4 admission rule. Enumeration has a strict cap and returns UNKNOWN when incomplete. Exactness is conditional, exponential, and limited to the fixed finite model and policy classes.

### 9.2 Frozen protocol and rejected primary hypothesis

The selection rule was recorded before a 20-seed pilot (20000–20019) and a 100-seed holdout (30000–30099). Every stream uses the inherited two-state, two-action, two-output generator, 16 action words, horizon four, and 48-episode A–B–A objective schedule. Four arms yield 3,840 pilot and 19,200 holdout rows: v4 B16, decision-regret B16, its untrained ablation, and the original optimistic full-history planner. All receive their own feedback and identical interaction horizons. Total CPU time is measured but unequal. The primary contrast is all-episode mean reward of the new trained controller minus v4 B16; intervals pair independent seeds rather than treating episodes as independent.

The proposed improvement is rejected. New mean return is 5.82708 versus 5.92167 for v4: difference  $-0.09458$ , paired 95%  $t$  interval  $[-0.16915, -0.02001]$ . Against full-history planning the difference is  $-0.11750$ , interval  $[-0.19182, -0.04318]$ . Late-A return falls from 5.80 to 5.72, with three seeds worse and 97 tied; its paired interval crosses zero. The largest observed anchor deficit increases from 4 to 10, still within the allowed 16. No protected-incumbent violations occur. Mean measured mechanics cost is approximately 1.90 times v4. Fewer paired-check nodes do not imply lower cost: the new arm adds about 5,957 model rollouts per stream for regret scoring.

| System                     | Mean return | Late A | Max. deficit | Seconds/stream |
|----------------------------|-------------|--------|--------------|----------------|
| Witness v4, B16            | 5.9217      | 5.8000 | 4            | 0.0359         |
| Regret probes, B16         | 5.8271      | 5.7200 | 10           | 0.0680         |
| Regret probes, no training | 5.8600      | 5.8000 | 10           | 0.0645         |
| Full-history planner       | 5.9446      | 5.8000 | 4            | 0.0466         |

**Table 2:** New frozen v5 holdout: 100 seeds, 48 episodes, four systems. Rewards range from zero to eight. Deficit is the worst completed-prefix loss relative to the initial anchor. Times cover measured per-stream decision/update mechanics, not setup or a dedicated timing experiment; concurrent host load can affect them.

Training also fails to show a clear benefit in this comparison: the trained minus untrained difference is  $-0.03292$ , interval  $[-0.09639, 0.03055]$ . This secondary interval is descriptive and uncorrected for multiplicity. The result does not contradict conditional incumbent retention; it shows that retention alone neither chooses useful probes nor ensures efficient learning. The pilot contains 20 distinct tables and the holdout 85, with 10 shared. The new holdout also shares seven tables with v4 development and 22 with its holdout. These are samples with replacement from only 256 labelled tables, not disjoint environments or evidence of new-domain transfer.

### 9.3 Complementary free probes defeat one-step scoring

An independent post-hoc counterexample isolates a completeness failure. There are four compatible worlds indexed by hidden bits  $(a, b)$ . An anchor returns 5 and reveals nothing. Two other actions also return 5 but reveal  $a$  and  $b$ , respectively. Two terminal decision actions return 10 or 0 depending on the parity  $a \text{ xor } b$ . Observations encode these outcomes; all four worlds are valid stationary reset models in a separately declared finite class. This construction uses more outputs/actions than the primary binary study and is identified as such in its executable artifact.

Initially the minimax regret is 5. Either free bit probe leaves parity ambiguous, so its immediate regret reduction is zero. The decision actions have positive regret reduction but a worst-case debit of 5 and are infeasible at budget zero. The greedy selector therefore repeats the anchor indefinitely. Nevertheless, executing the two free probes identifies parity and permits a guaranteed improvement of 5 without any exploration loss. The reproducer executes both the stall and the successful probe sequence; it is a finite implementation test and written induction, not an additional Lean theorem.

**A separately diagnosed admission plateau.** Three frozen-holdout seeds stall after the sum of minimax regrets reaches zero. A common optimal candidate exists, but it ties the incumbent in at least one compatible model, so the strict  $L > 0$  promotion rule refuses it.

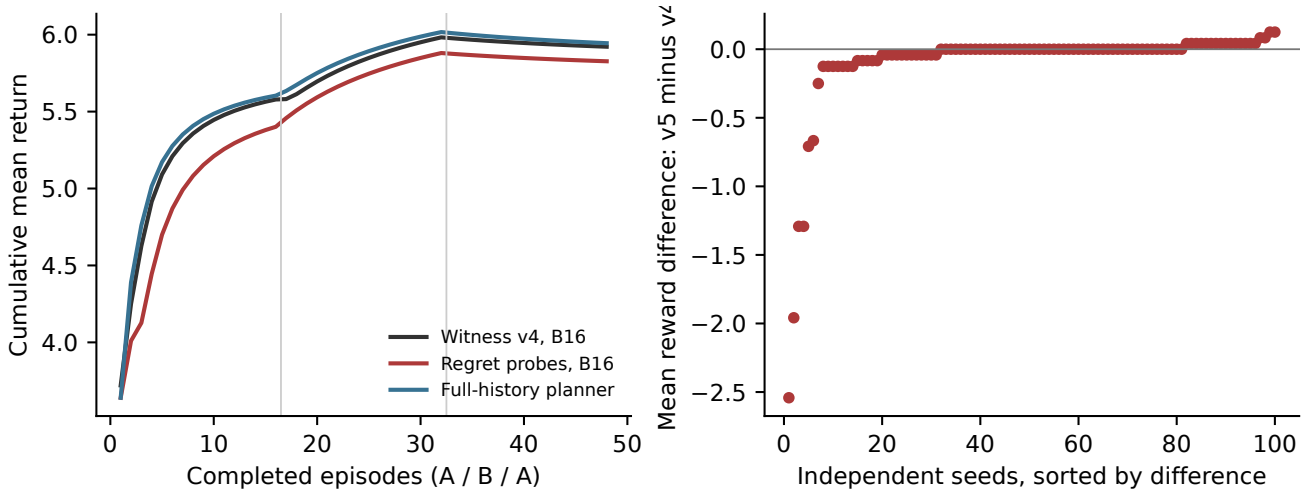
Every probe also has zero regret-reduction score. A separate weak-dominance closure promotes a candidate only when exact comparison gives  $L \geq 0$  and  $U > 0$ , preserving the conditional retention premise. This is a post-hoc repair, not the frozen method.

An exhaustive development diagnostic evaluates all 256 labelled tables under one proposer initialization per table, four arms and 48 episodes: 49,152 rows. The repair improves the original v5 mean by 0.05453, improves ten tables and worsens none, and repairs four late-return deficits in that diagnostic. It still trails v4 and the full-history planner (mean 6.25326 versus 6.26123 and 6.29541) and is slower. This finite schedule census supports the diagnosed repair only; it supplies no holdout confidence interval or guarantee for other schedules, model classes, or random initializations. It does not resolve the complementary free-probe obstruction above.

**Successor hypothesis.** A bounded, adaptive two-probe plan can value complementary information that Equation (11) misses. For each first outcome, choose a second probe using that branch’s updated cover, and maximize the worst-branch terminal reduction. Charge every executed probe separately using the original ledger; simulated outcomes supply planning information, never training evidence. A cheaper baseline simply falls back to v4’s trace-count rule when  $G = 0$ . Neither approach is entitled to a reward guarantee from uncertainty contraction. The first kill test is failure to solve the complementary-probe example at zero budget; the substantive kill test is no future reward advantage over both v4 and the cheaper fallback at equal total compute. The existing holdout is now consumed for design purposes. Any tuned successor evaluation must be labelled post-hoc or use a genuinely new task family/protocol.

## 10 Actual local-model inference and native integration

A first actual local-model smoke run used the already available 27B Q3 GGUF model behind a loopback llama.cpp server, with temperature zero, a 96-token output cap, and one call per episode for each model arm. The arms are raw executed-history prompting, the same



**Figure 3:** Frozen v5 result from raw episode records. Left: mean cumulative reward, including all exploratory episodes. Right: per-seed mean differences sorted for visibility; the seed is the statistical unit. The public objective changes at 16 and 32 completed episodes. Underlying CSVs accompany the figure.

prompt plus the v4 checker and online proposer, and a checker-based static-library control with no model calls. Programs are frozen before execution; invalid output consumes the call and triggers an explicitly recorded fallback, with the rejection retained. No offline retraining or paid API is used.

The first fixed seed, 41000, produced 36 episodes and 24 model calls with no invalid outputs. All arms returned 5.3333 on average. Evaluator-only enumeration then found every program had identical reward in that world, so this result has no learning headroom. It establishes only executable inference and accounting. A subsequent four-consecutive-seed exploratory extension (41001–41004) produced 144 episodes and 96 model calls. Mean returns were 3.9583 for raw history, 5.5417 for checked history, and 5.0000 for the static-library control. Both model arms used 13,540 input and 576 output tokens, with respectively 141.91 and 128.51 seconds of observed model-call latency. Both stages together contain 120 actual calls, all with valid output and complete usage records. The static-library control still updates the small auxiliary ranker; only its candidate library and absence of LLM calls are static.

These are five environments, not 60 independent trials per arm. No stateless LLM control, native task, or powered retention evaluation ran. The symbolic checker has a strong finite-class prior, more persistent state, extra CPU work, and online auxiliary training. No LLM weights change. The pilot supports integration feasibility and motivates a native experiment; it does not isolate memory quality, parameter learning, or cost-matched superiority. Initial model loading took about 77.7 seconds and is excluded from call timings. Host/GPU memory values are environment observations, not isolated peak-memory benchmarks.

An independent review found misleading malformed-

output metadata and incomplete usage-field handling in the pilot driver. Neither branch was exercised in these runs: every completion was valid and every usage record complete. Future-run code and regression tests are corrected; the exact executed source, original results, and review are preserved. For invalid future output, raw history uses the anchor while the checked arm skips registration and uses its existing controller, with no extra call. This distinction is now explicit.

The native integration targets the official CL-Bench interface at commit `5f8c50eb1e84b2eda2ef4faff757dfc812a0ea26`. Its feedback semantics are material: the runner separately delivers observations; duplicating them through query handling would change the ICL baseline. A task may legally reveal a correct answer in post-action feedback; that is different from reading the evaluator’s answer records before acting. Native reward and each system’s stateful-stateless gain must remain separate. A bridge and interface tests do not supply a native benchmark result, an empirical retention result, or a valid finite-state abstraction for database drift. Those remain the next experimental obligations.

Twenty optional-runtime tests pass, including eight tests executing the actual pinned upstream ICL/runner and SQLite task; twelve dependency-free contracts also belong to the 401-test standard suite. The native transport is scripted for these checks, and the SQLite database/questions are fixtures. Its exact certificate metadata is always UNKNOWN. The bridge is a baseline and audit foundation, not an implemented native Witness adaptation algorithm. The unmodified official benchmark dataset and a live native model transport remain unexecuted.

## 11 Limitations and decisive next experiments

**Where the guarantee breaks.** Consistency with observed traces does not validate  $K$ , stationarity, reset semantics, deterministic feedback, or output sufficiency. A point estimate or underspecified class can certify a change that fails in a larger compatible world. Hidden drift can invalidate a previous certificate before its first counterexample arrives. Public objectives do not solve hidden task routing. The fixed reset-value theorem is weaker than v3’s supplied-model all-state claim and much weaker than broad capability retention.

There is also an unavoidable information–risk tradeoff within some classes. Two worlds can make the anchor observationally identical while a distinguishing action has opposite reward consequences. Zero-risk execution cannot both retain the anchor in the unfavorable world and guarantee learning the favorable one. The right target is therefore not an unconditional monotonicity slogan, but a useful, explicit and affordable tradeoff. Independent new information can also require growing total storage; our archive does not bypass that cost.

**Native falsification.** The next experiment must first acquire a valid abstraction from a real, resettable tool environment without using hidden tables or task answers. Then evaluate compatible CL-Bench domains and AgentCL streams with the same fixed backbone, permissions, generation settings and total compute budget. Include raw-history ICL, a strong memory/skill baseline, an evolving harness, point-estimate models, and removals of checking, training and informative probes. Count proposal sampling, gradient updates, checker work, retrieval, replay, resets, retries and environment interactions. An oracle-representation arm is diagnostic only.

A concrete proposed success criterion is at least five percentage points of native normalized learning gain over the strongest cost-matched baseline, positive lower paired 95% bound, and no protected old-task success loss larger than two percentage points. These are research targets, not achieved results. Reject the scalable hypothesis if no sufficient small abstraction survives held-out continuation tests, if over 90% of candidate checks return UNKNOWN on the selected native tasks, if apparent gains vanish at matched compute, or if cost exceeds the strongest baseline without the predeclared gain.

**Resources and execution design.** The current simulator requires CPU only. The first local inference pilot uses an existing 27B Q3 model with 48 GPU-offloaded

layers on an RTX 5070 Ti, 4,096 context tokens and a single server slot. A larger native run needs a measured fixed-backbone inference budget and, where required, a Docker host. Measure native pilot throughput before assigning GPU-hours; the small synthetic run does not establish native task throughput. Retain full provenance and measure actual peak host/GPU memory rather than payload estimates.

A possible systems extension groups partial-transition work by time and unresolved slot in structure-of-arrays queues, with exact residual IDs and checked reductions. CPU parallelism should precede a GPU rewrite for tiny irregular checks. A proof cache must include immutable programs, objective, reset contract and evidence ancestry; enlarging the class invalidates reuse. These are unimplemented engineering hypotheses. A GPU cannot remove exponential model ambiguity, and an approximation cannot inherit exact guarantees without a separate soundness argument.

## 12 Conclusion

Witness-CL v0.5 learns compatible hidden-state models from its own executed traces and uses them to protect a behavioral incumbent independently of a mutable proposer. Version 5 adds checked abstract Lean statements, real local inference, and a falsified regret-probe heuristic with an explicit complementary-information counterexample. These narrow results separate retention from learning efficiency. It does not yet supply the requested general stateful benchmark winner. The remaining research obligation is concrete: learn sufficient continuation abstractions cheaply enough that verified reuse beats full-history learning at equal cost, without hiding class misspecification or old-skill loss behind the word “memory.”

## Reproducibility and disclosure

Source, tests, raw episode files, result generation, original Git history, initial and successful Lean build evidence and paper sources are included in the private research repository. The main reproduction targets are `make test`, `make cpp-check`, `make latent-heldout`, `make latent-diagnostics`, and `make paper`. The v5 local-model pilot was executed; its small scope is explicit. This is an AI-assisted research draft prepared for Samuel Mausberg, with author review and independent replication pending. The native learning problem remains open.

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